

Loyola Marymount University

CMSI 3300 Artificial Intelligence

Fall 2024

Professor Lanyu Shang

Announcements

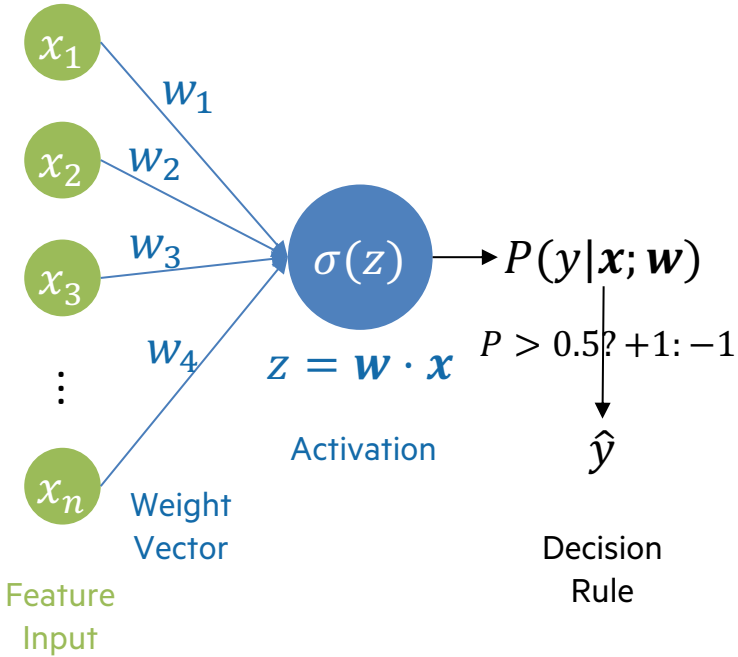
- Classwork 6
  - Due next **Tuesday** 11/26 @5pm
- Homework 4
  - Due Friday 12/6 @5pm

This Lecture

- Neural Networks
  - Artificial neural networks
  - Feed-Forward neural networks

Last Lecture

- Logistic Regression



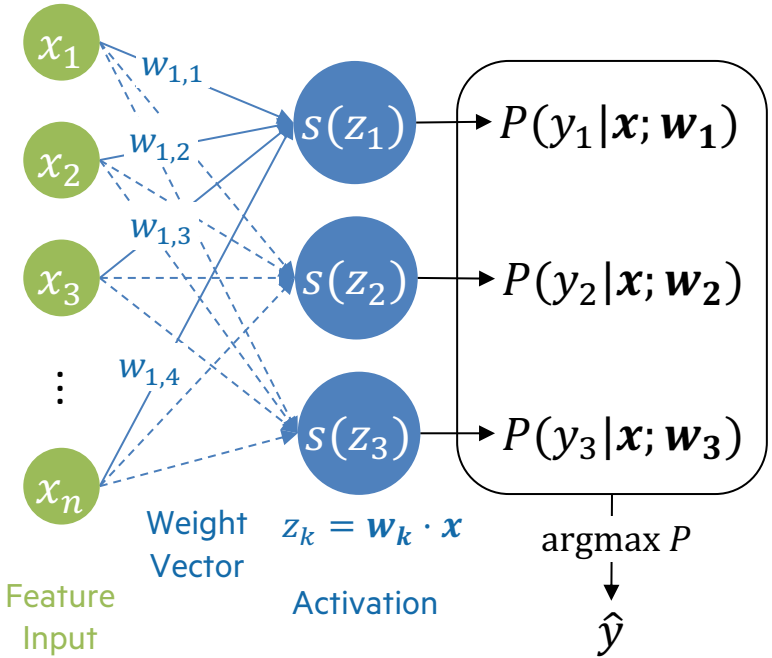
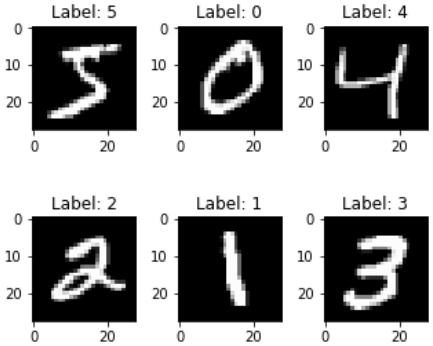


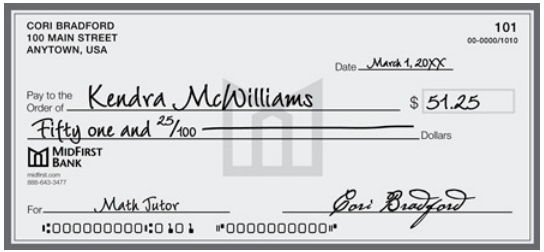
Image Classification Problem

- Hand-written Digit Classification
  - MNIST dataset
    - Numbers from 0-9
    - Training/Test: 60K/10K

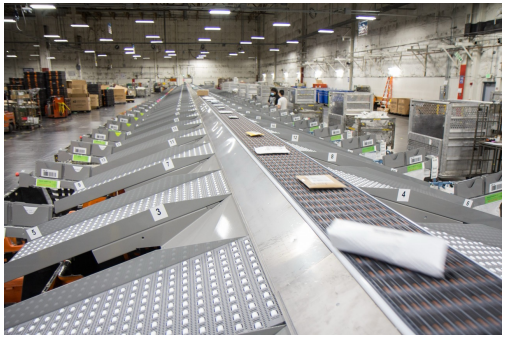
What type of learning is it?



- What are the real-world applications?



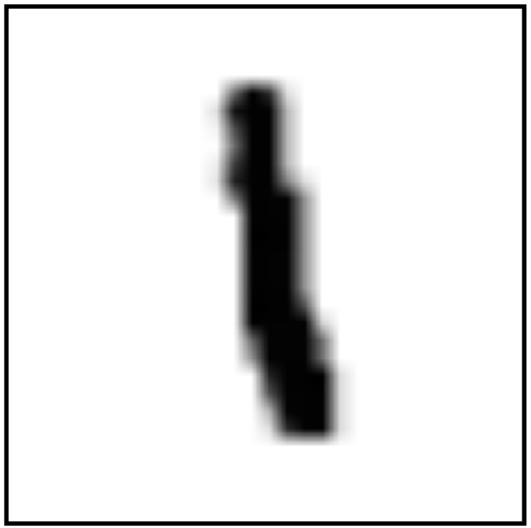
Bank Check Processing



USPS Mail Sorting

MNIST Dataset

- 28x28 pixel images
- Grayscale: 0 (black) to 255 (white)

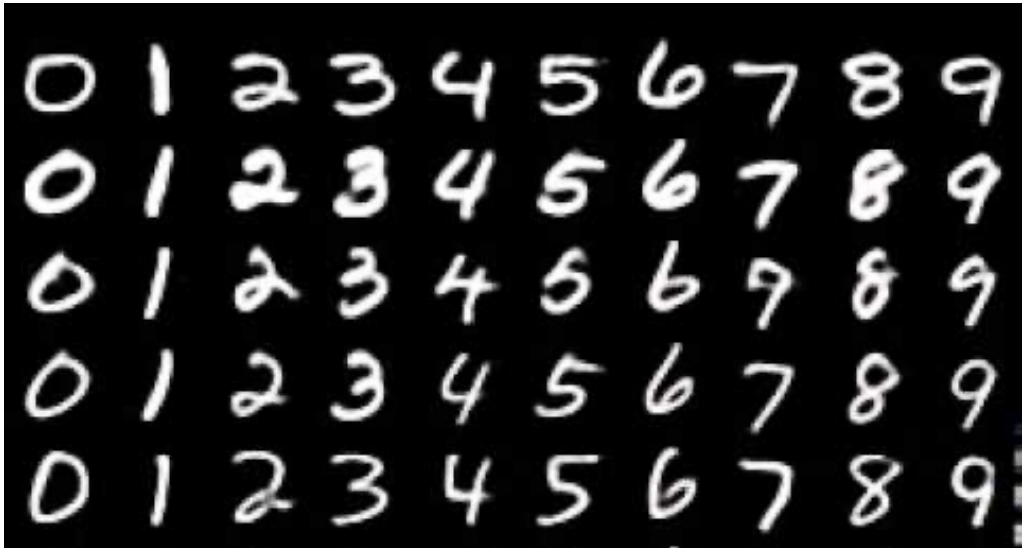


$\approx$

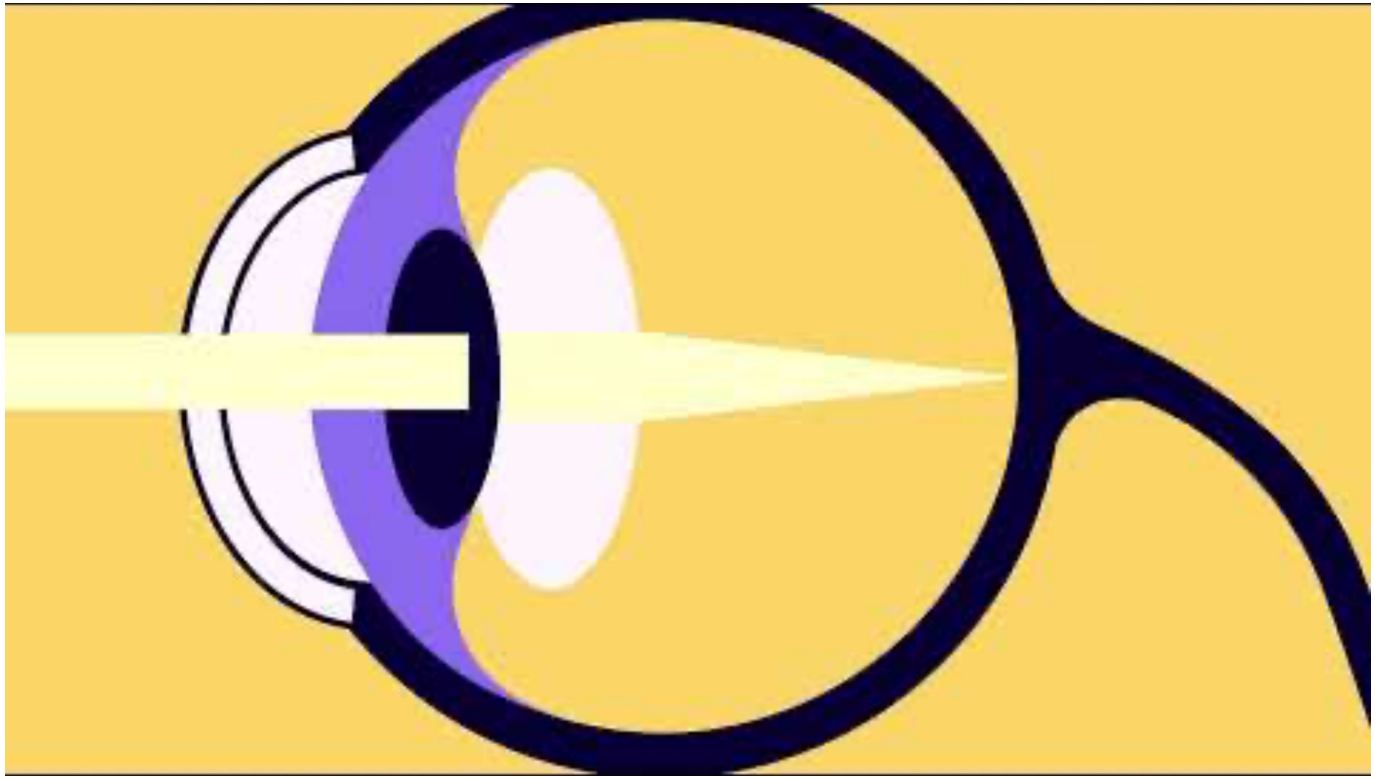
$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & .6 & .8 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & .7 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & .7 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & .5 & 1 & .4 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & .4 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & .4 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & .7 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & .9 & 1 & .1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & .3 & 1 & .1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

Classification

- If we want to build a logistic regression model, what are the **features** to consider?



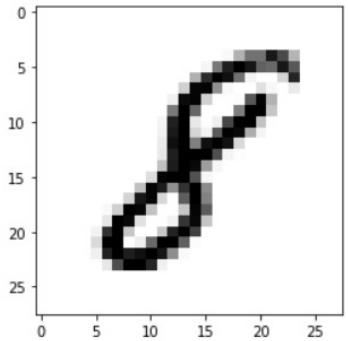
Human Visual System



Video Link: https://www.youtube.com/watch?v=TY1giZgddAs

Human Visual System

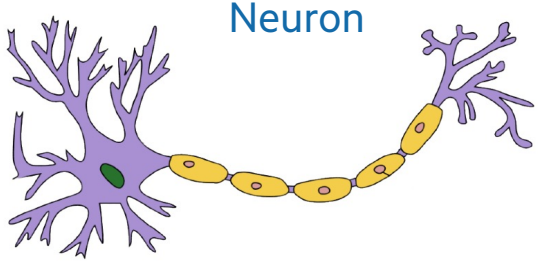
- Input
  - Primitive visual features
    - Basic, low-level characteristics that can be extracted from raw data (e.g., image pixels)
- Output
  - Classification
    - Objects/Digits/…
- How to implement it?



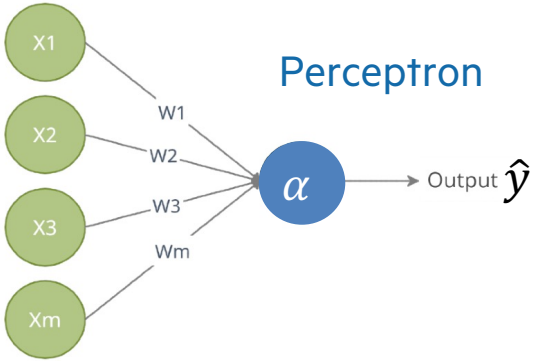
What are the dimensions of the input and output?

Label: 8

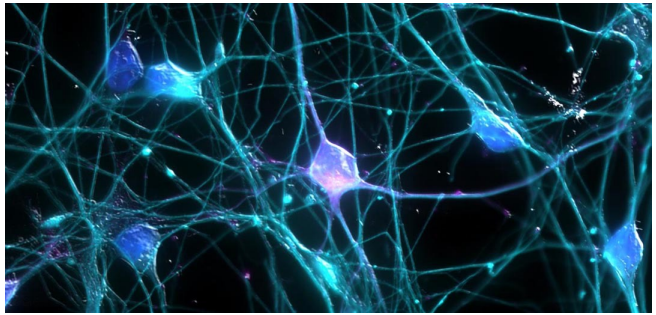
(Artificial) Neural Network



Neuron



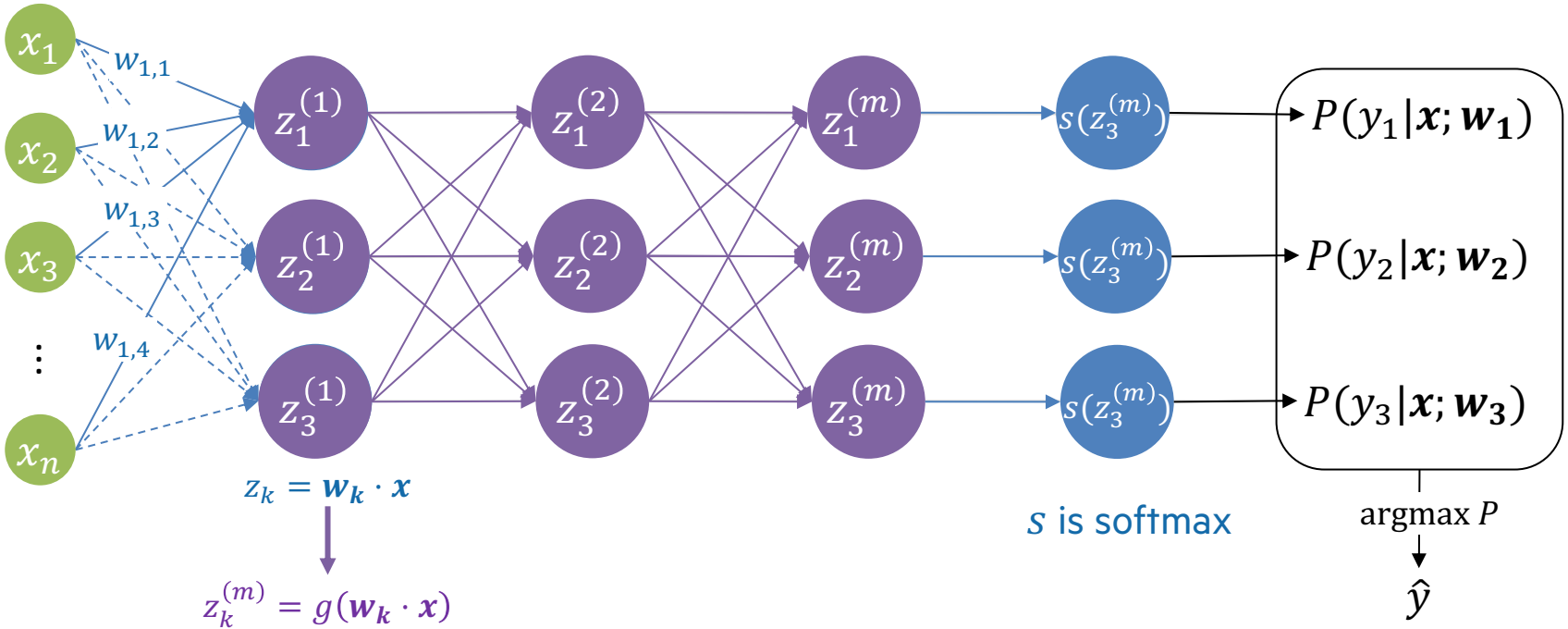
Perceptron



Neurons in Brain

Connected Neurons (Neural Network)

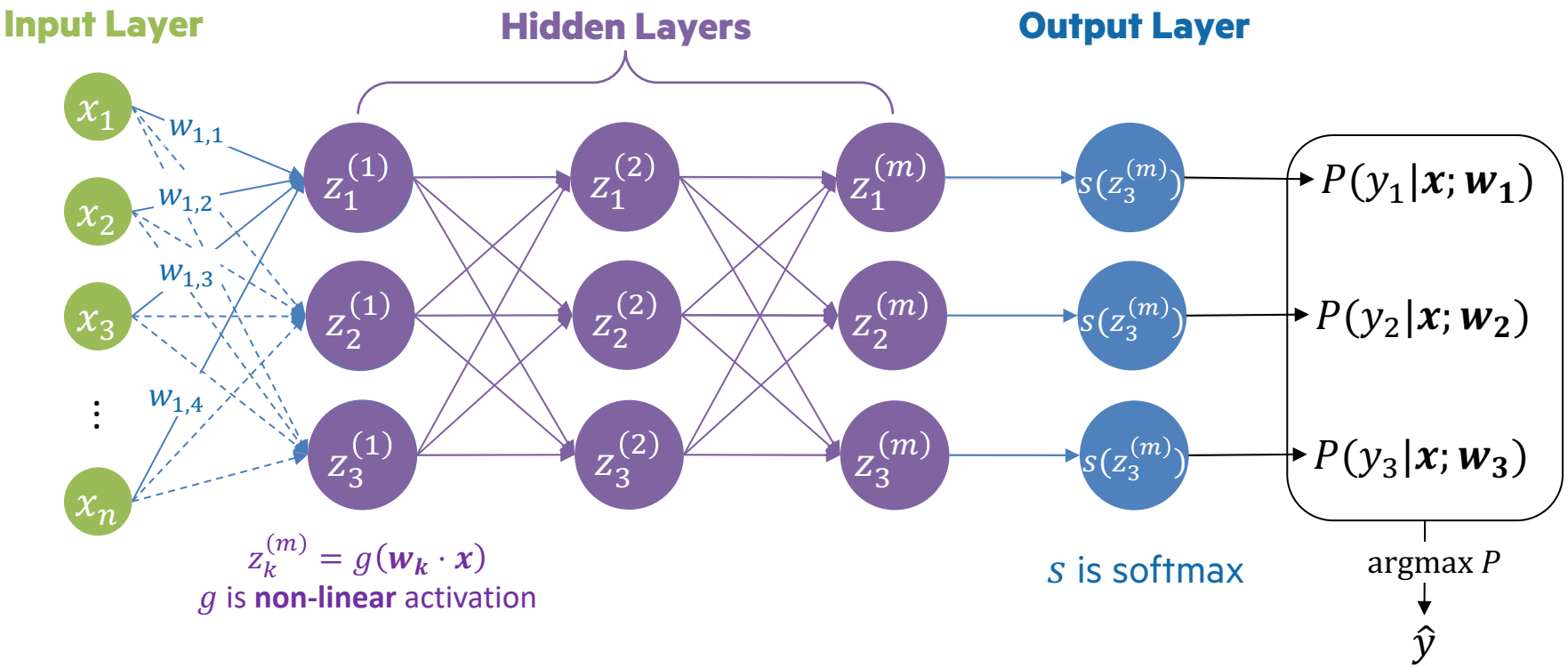
Feed-Forward Network



Feed-Forward Network

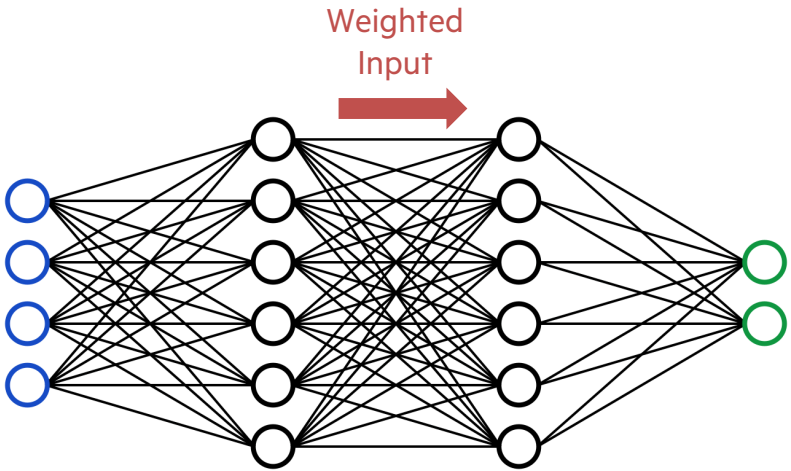
- Input layer (leftmost)
  - Raw-input
  - E.g., pixels of an image, words from an email
- Output layer (rightmost)
  - One output for each class $y \in Y$ (e.g., softmax likelihoods)
- Hidden layers (between the above)
  - Learn higher order features from the raw input
  - Can have 1 or more hidden layers

Feed-Forward Network

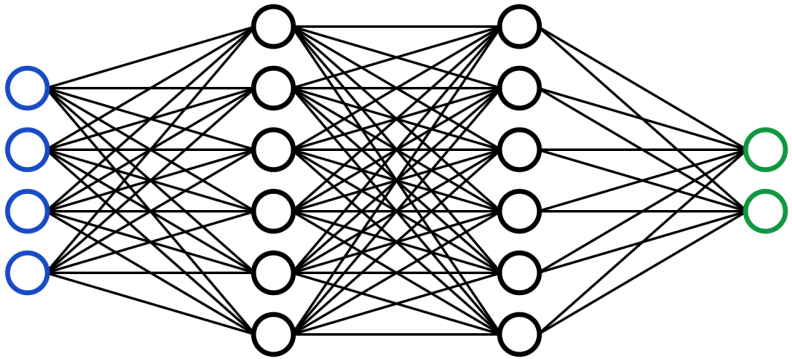


Feed-Forward Network

- In a densely connected network, all units/neurons in previous layer serve as **weighted input** into next
- Each edge has its own **weight** $w_{k,i}^{(m)}$
- Each neuron has its own **non-linear activation** (e.g., sigmoid)

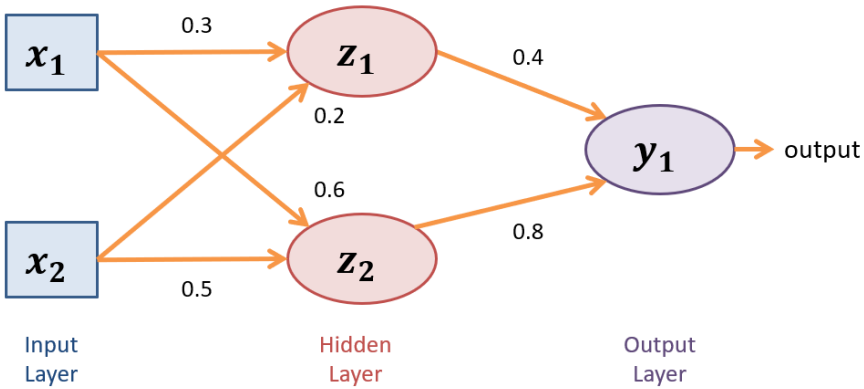


DIY Neural Networks



Example

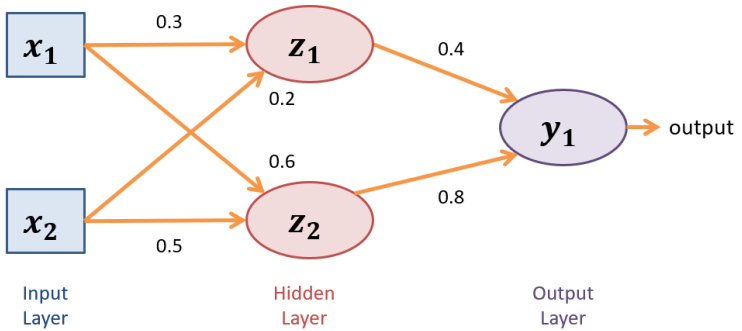
- Input: $< x_1, x_2 >$
- 1 hidden layer with 2 neurons / units $\{z_1, z_2\}$
- Output: y_1
- Weights as shown in the figure
- Activation: sigmoid of weighted sum of input
  - $z = \sigma(\boldsymbol{w} \cdot \boldsymbol{x}) = \frac{1}{1+e^{\boldsymbol{w} \cdot \boldsymbol{x}}}$



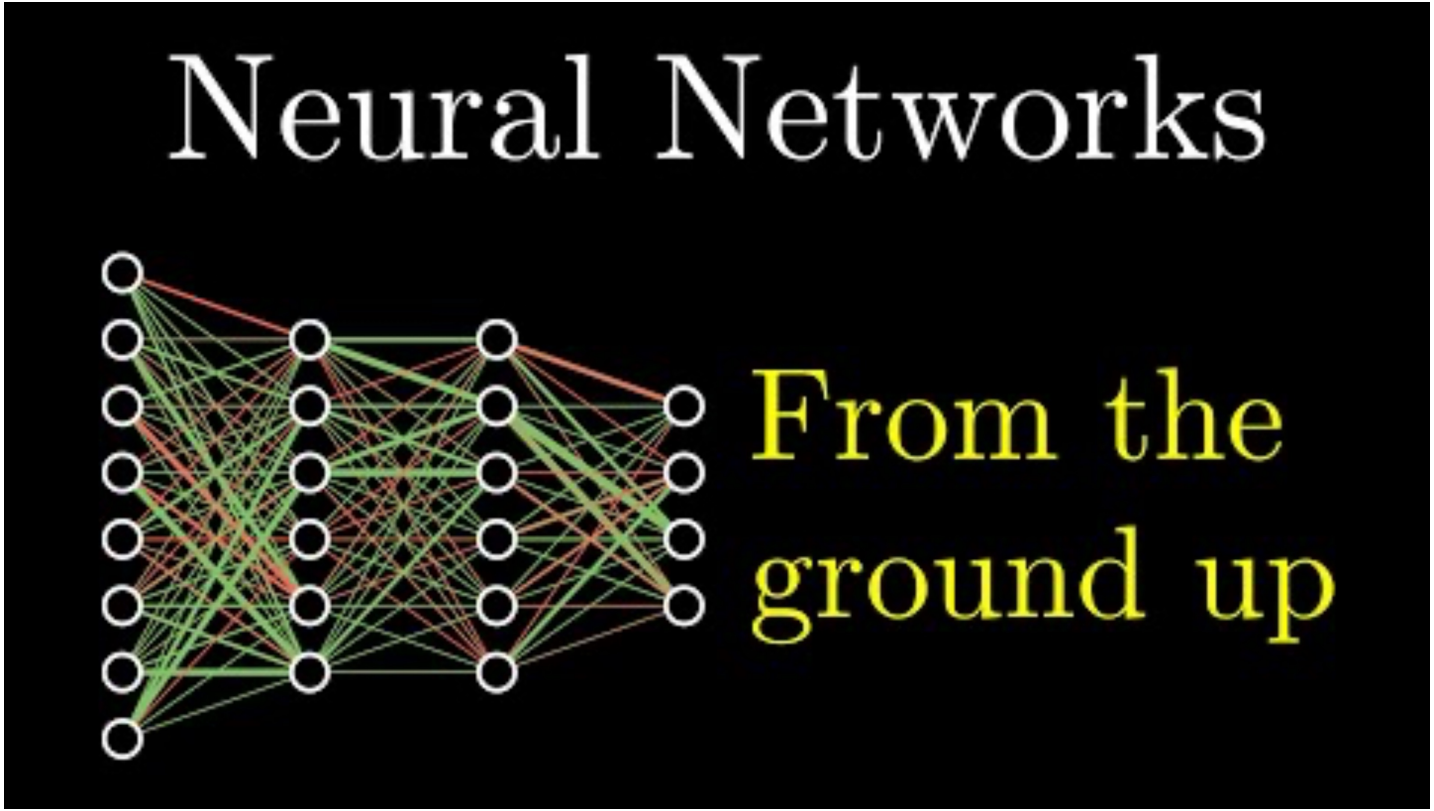
Question: Given $x_1 = 1, x_2 = 2$, compute the output of this network at y_1

Example

- $z_1 = \sigma(1 * 0.3 + 2 * 0.2)$
  ≈ 0.67
- $z_2 = \sigma(1 * 0.6 + 2 * 0.5)$
  ≈ 0.83
- $y_1 = \sigma(0.67 * 0.4 + 0.83 * 0.8)$
  ≈ 0.72



MNIST Example



Video Link: https://www.youtube.com/watch?v=aircAruvnKk